Almost sure parameter continuity of subcritical exponential metrics in every dimension

Abstract

Let $d \geq 2$ and let h be a whole-space log-correlated Gaussian field. We give a common proof of almost sure parameter continuity for the subcritical exponential metrics associated with h, including the Euclidean metric at parameter zero. The metrics are normalized by the median distance between two fixed unit-separated points. In dimension two the fixed-parameter metric characterization is the theorem of Gwynne–Miller. In higher dimensions the strong characterization is the standing project input specified below, as in the higher-dimensional source manuscript. The argument uses one quantitative routing construction in every dimension, one comparison estimate, and one construction of the continuous modification. A short Gaussian energy argument supplies the qualitative Euclidean limit from the same one-metric estimates used in the comparison. The resulting family consists simultaneously of proper geodesic metrics which induce the Euclidean topology.

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1 Introduction

Let $d \geq 2$. We consider a whole-space log-correlated Gaussian field h on $\mathbb{R}^d$, with logarithmic covariance coefficient one and with its average on the unit sphere equal to zero. When $d = 2$, this is a whole-plane Gaussian free field. In higher dimensions it is the log-correlated field, not the ordinary Gaussian free field. For $0 < \gamma < \sqrt{2d}$, let D_h^γ be its strong subcritical exponential metric, with the normalization

$$\operatorname{med} D_h^\gamma(0, e_1) = 1\,. \tag{1.1}$$

Here and below the median is the lower median. Set $D_h^0(x, y) = |x - y|$. All parameters are coupled through this one field.

Fixed-parameter input. For $d = 2$, we use the metric characterization of Gwynne–Miller [1]. For $d \geq 3$, we use the strong metric characterization stipulated in the higher-dimensional source manuscript [11], namely the project theorem [10]. This includes locality, Weyl scaling, exact similarity covariance, and the identification of the normalized LFPP limit. It is an external input, not a consequence claimed here of the published tightness theorem alone. Thus the higher-dimensional conclusion below has exactly the same standing input as [11]. The uniform one-metric estimates used from the two source manuscripts are recorded precisely in Proposition 2.1. Neither source continuity theorem is used as an input.

Theorem 1.1. *Under the fixed-parameter input just described, the metrics D_h^γ admit a field-measurable modification $\widetilde{D}_h^\gamma$ such that, almost surely,*

$$[0, \sqrt{2d}) \ni \gamma \longmapsto \widetilde{D}_h^\gamma \in C_{\mathrm{loc}}(\mathbb{R}^d \times \mathbb{R}^d) \tag{1.2}$$

is continuous. On the same event, every $\widetilde{D}_h^\gamma$ is a proper geodesic metric which induces the Euclidean topology, and $\widetilde{D}_h^0(x, y) = |x - y|$. At each deterministic γ, the modified metric agrees almost surely with the prescribed metric normalized by (1.1).

The assertion concerns a modification, not an uncountable intersection of the events on which separate versions are defined. In particular, agreement with a metric at every deterministic parameter does not, by itself, rule out a random parameter at which a continuous pseudometric family degenerates. We address this issue separately in Section 7.

1.1 Outline of the proof

The main estimate bounds increments in the Weyl parameter ξ. On each compact subcritical interval, sufficiently close parameters satisfy a quantitative comparison with an error probability which decays exponentially in a scale-depth variable. The permitted parameter spacing for a fixed initial accuracy can be exponentially small in a power of that accuracy. This is enough: the failure probability after amplification is much smaller than the cardinality of the parameter grids we use.

The comparison has two parts. A finite collection of Cameron–Martin shifts produces a fixed-confidence comparison. Each shift discounts a connected network and forces a primary geodesic to traverse a portion with a cheaper secondary connection. Counting the possible shifts improves an assumed comparison coefficient. Separated annuli then amplify this fixed-confidence comparison. The different similarity factors of the two metrics are retained throughout.

The geometric construction is the same for all $d \geq 2$. A quantitative retraction onto a controlled arc bounds the holes in a cubical neighborhood of that arc. The exterior cubical grid remains

connected after deleting any one vertex. Menger's theorem therefore supplies two disjoint access paths, with either pairing of their endpoints. This replaces both the planar routing argument and the higher-dimensional general-position argument.

For positive parameters, nested grids and Borel–Cantelli give a continuous family. Near zero, the one-metric estimates give a uniform bound on the logarithmic energy of geodesic occupation measures. Gaussian Poincaré then proves convergence in probability to the Euclidean metric. A coarse grid, one intermediate comparison, and the same fine grids upgrade this to almost sure convergence through all real parameters. No first-order expansion or quantitative Euclidean-limit theorem is needed.

1.2 Notation and normalization

Write $B_r(z)$ for a Euclidean ball and $B_r = B_r(0)$. The average of h on $\partial B_r(z)$ is $h_r(z)$. Internal distances in an open set U are denoted by $D(x, y; U)$. For a continuous function of two spatial variables, set

$$\|F\|_R = \sup_{x,y\in\overline{B}_R} |F(x, y)| .$$

The parameter correspondence and the scaling exponent are

$$Q_d(\xi) = \frac{d}{\gamma} + \frac{\gamma}{2}, \qquad \xi = \frac{\gamma}{d_{\gamma,d}}, \qquad \phi(\xi) = \xi Q_d(\xi) . \tag{1.3}$$

Here $d_{\gamma,d}$ is the almost sure Hausdorff dimension of $(\mathbb{R}^d, D_h^\gamma)$. Only the parameter correspondence in (1.3), not a formula for this dimension, is used. The map $\gamma \mapsto \xi$ is an increasing homeomorphism from $[0, \sqrt{2d})$ onto $[0, \xi_c(d))$; we set $\xi(0) = 0$ and $\phi(0) = 1$. We use the parameter correspondence supplied with the fixed-parameter characterization and exponent bounds; see [2, 5, 10].

We first use mean-log normalization. For the unit-sphere pin, define

$$m(\xi) = \mathbb{E} \log D_h^{\gamma(\xi)}(0, e_1), \qquad \rho_h^\xi = e^{-m(\xi)} D_h^{\gamma(\xi)}, \qquad \rho_h^0(x, y) = |x - y| . \tag{1.4}$$

The expectation is finite by Proposition 2.1. We do not assume continuity of m. The return to median normalization is made only after the continuous family has been constructed.

The field-change argument uses the radius-ten pin. If $g = h - h_{10}(0)$, then

$$\rho_h^\xi = e^{\xi h_{10}(0)} \rho_g^\xi, \qquad \mathbb{E} \log \rho_g^\xi(0, e_1) = 0 . \tag{1.5}$$

The deterministic mean-log normalization is held fixed under field changes. Exact similarity covariance gives

$$\rho_h^\xi(z + rx, z + ry) = S_\xi(r, z) \rho^\xi_{h(z+r\cdot)-h_r(z)}(x, y), \qquad S_\xi(r, z) = r^{\phi(\xi)} e^{\xi h_r(z)}, \tag{1.6}$$

$$\log \frac{S_\eta(r, z)}{S_\xi(r, z)} = (\phi(\eta) - \phi(\xi)) \log r + (\eta - \xi) h_r(z) . \tag{1.7}$$

There is an identical formula for the radius-ten pin, with $h_{10r}(z)$ in place of $h_r(z)$.

Throughout the comparison, fix $J = [0, b] \Subset [0, \xi_c(d))$. Constants may depend on d, J and on the displayed fixed spatial domains, but not on the lower parameter bound a, which is introduced later. Different occurrences of C, c may have different values. An estimate described as polynomial in v^{-1} has a fixed exponent depending only on these fixed data. No uniformity as $d \to \infty$ is asserted.

2 One-metric estimates and the Euclidean limit

We separate the spatial estimates for a single deterministic parameter from the comparison of two parameters. This is important both for the dependence of constants near zero and for the common-event construction at the end.

Proposition 2.1 (Uniform one-metric estimates)**.** *For each $J = [0, b] \Subset [0, \xi_c(d))$, there are constants $p, p_{\mathrm{esc}} > 0$, $0 < \sigma < 1 < \beta$, and $C, A < \infty$ with the following properties. For each deterministic $\xi \in J$ and $R \geq 1$, there is $H_{R,\xi} \geq 1$ such that*

$$\sup_{\xi \in J} \mathbb{E} H_{R,\xi}^{p} \leq C(1+R)^{A}, \tag{2.1}$$

$$\operatorname{diam}_{\rho_h^{\xi}}(B_s(z); B_{2s}(z)) \leq H_{R,\xi} s^{\sigma}, \qquad z \in B_R, \quad 0 < s \leq 1, \tag{2.2}$$

$$\rho_h^{\xi}(x, y) \geq H_{R,\xi}^{-1} |x - y|^{\beta}, \qquad x, y \in \overline{B}_R. \tag{2.3}$$

Fixed compact annular crossings have inverse moments of every prescribed finite order, uniformly in ξ. Fixed buffered internal diameters have a uniform positive moment. Fixed nonzero point distances have uniformly bounded logarithmic moments of every finite order.

For each fixed $K > 1$, except on an event of probability at most $C_K S^{-p_{\mathrm{esc}}}$, there is a competitor for every pair in B_R whose length is less than K^{-1} times the cost of a mandatory crossing to B_{SR}^{c}. Here $S \geq C_K$, and the comparison uses the same metric for the competitor and the crossing. Every fixed-parameter metric is proper and geodesic. All assertions hold under either fixed sphere pin.

The function ϕ is Lipschitz on J, and

$$\inf_{\xi \in J} \left(\phi(\xi) - \sqrt{2d}\, \xi \right) > 0 \,. \tag{2.4}$$

Source verification. For $d = 2$, the metric estimates are the conclusions of [12, Section 2, in particular Proposition 2.8]; they follow there from the planar estimates of [6], inverse-crossing amplification, and the metric characterization [1]. For $d \geq 3$, they are [11, Lemma 2.1 and Appendix A, Proposition A.3], transferred to the continuum by the fixed-parameter identification [10]. Appendix A of [11] proves the estimates uniformly over the parameter interval; it does not use that manuscript's almost sure continuity theorem. These source propositions are used as one-metric results, not as substitutes for the two-parameter estimate proved below. Its normalization is exactly (1.4). These are the only metric-estimate results imported from the two continuity manuscripts.

For clarity, the exponent regularity does not require continuity of the metric family. On a fixed box, the logarithm of a cutoff distance satisfies

$$|\log L_n^{\xi} - \log L_n^{\eta}| \leq |\xi - \eta| \sup |h_n| \,.$$

The expected supremum is at most $C(n+1)$. The logarithmic Gaussian concentration error between its mean and median is $O(b\sqrt{n})$. Dividing by $n \log 2$ and using the defining distance exponent gives the Lipschitz bound for ϕ. The subcritical relation $Q_d(\xi) > \sqrt{2d}$ and continuity, with $\phi(0) = 1$, give (2.4). The same estimates compare the logarithmic normalizers, so no parameter-continuity statement is hidden in this calculation. Changing the sphere pin multiplies the metric by a centered Gaussian exponential; Hölder's inequality, with a smaller positive moment order when necessary, preserves all the stated bounds. □

2.1 A short proof of the marginal limit at zero

The following argument uses only Proposition 2.1, Weyl scaling, and the Gaussian structure. In particular, it removes the dependence of the higher-dimensional endpoint argument on the first-order fluctuation manuscript.

Lemma 2.2. *For every $R < \infty$, under either fixed sphere pin,*

$$\|\rho_h^\xi - |\cdot - \cdot|\|_R \longrightarrow 0 \quad \text{in probability as } \xi \downarrow 0\,. \tag{2.5}$$

Moreover, for the unit-sphere pin and all sufficiently small ξ,

$$\mathbb{E}\big[\log \rho_h^\xi(0, e_1)\big]^2 \le C\xi^2\,. \tag{2.6}$$

Proof. Put $F_\xi = \log \rho_h^\xi(0, e_1)$, so $\mathbb{E}F_\xi = 0$. Choose K with $Kp > A + 1$ and set

$$\mathcal{H}_\xi = 1 \vee \sup_{n\ge 0} 2^{-Kn} H_{2^{n+2},\xi}\,.$$

Equation (2.1) gives $\sup_\xi \mathbb{E}\mathcal{H}_\xi^p < \infty$. Let $T_\xi \ge 2$ be the smallest dyadic radius which contains every geodesic from 0 to e_1. Relative escape gives $\mathbb{P}[T_\xi > t] \le Ct^{-c}$, after changing constants. In particular, all moments of $\log T_\xi$ are uniformly bounded.

For any such geodesic, parameterized at constant metric speed on $[0, 1]$, let ν be its occupation probability measure. All visits to a Euclidean ball occur between its first and last visits. The intervening segment is a geodesic, so (2.2), followed by a fixed local chain, gives

$$\nu(B_r(z)) \le \min\{1, B_\xi r^\sigma\}, \qquad B_\xi = C\mathcal{H}_\xi(1 + T_\xi)^{K+1} e^{-F_\xi}, \qquad 0 < r \le 1\,. \tag{2.7}$$

We may replace B_ξ by $1 \vee B_\xi$. The covariance kernel for the pinned field is $-\log|x-y|$ plus two one-variable terms and a constant. On $B_T \times B_T$, those terms have absolute value at most $C\log(2+T)$. Thus the squared Cameron–Martin dual norm of ν is at most

$$\begin{aligned}\|\nu\|_{\mathcal{H}^*}^2 &\le C\log(2 + T_\xi) + \iint \log^+ \frac{1}{|x-y|}\,\nu(dx)\nu(dy)\\ &\le C\big(1 + \log \mathcal{H}_\xi + \log(2 + T_\xi) + (F_\xi)_-\big) =: CU_\xi\,.\end{aligned} \tag{2.8}$$

The second inequality follows by integrating $\min\{1, B_\xi e^{-\sigma t}\}$ over $t \ge 0$. Proposition 2.1 implies $\sup_\xi \mathbb{E}U_\xi < \infty$.

Here is the Gaussian differentiation step. Choose an orthonormal Cameron–Martin basis with bounded smooth pinned representatives, obtained by Gram–Schmidt from a dense collection of smooth functions modulo constants. Along each basis direction f, Weyl scaling gives a Lipschitz representative of $F_\xi(h + tf)$ as a function of t. If the Weyl identity is initially stated separately for deterministic shifts, Fubini supplies this linewise representative; Gaussian absolute continuity then identifies its weak derivative at almost every field. At a point of differentiability, an unperturbed geodesic is a competitor for both signs of t, giving

$$F_\xi(h + tf) - F_\xi(h) \le \log \int e^{\xi t f}\, d\nu = \xi t \int f\, d\nu + O(t^2)\,.$$

Dividing separately by positive and negative t identifies the derivative. Fubini and a countable intersection give $\|\nabla F_\xi\|_{\mathcal{H}}^2 \le C\xi^2 U_\xi$. The logarithmic moment bound puts F_ξ in L^2; the coordinate

derivative bound, by finite-dimensional approximation and closure, puts it in the Gaussian Sobolev space. Gaussian Poincaré proves (2.6).

For fixed distinct x, y, exact scaling expresses $\log \rho_h^\xi(x,y)$ as $\phi(\xi)\log|x-y|$, a fixed-variance Gaussian multiplied by ξ, and a variable with the law of F_ξ. Hence it converges in probability to $\log|x-y|$. On a finite Euclidean net this holds simultaneously for every pair. The tight spatial upper bound (2.2) controls the interpolation error by a fixed power of the mesh. Letting first $\xi \downarrow 0$ and then the mesh tend to zero proves (2.5). The pin correction in (1.5) tends to one and gives the assertion for the other pin. □

3 Local comparison and annular amplification

We now compare $D = \rho_h^\xi$ and $E = \rho_h^\eta$. Put $\tau = |\xi - \eta|$. Until the continuous modification is constructed, the parameters in every probability estimate are deterministic.

3.1 Gaussian separation and physical units

The local events below are invariant under adding a field constant after their metric units have been removed. Their Gaussian observations are therefore zero-mass observations supported in fixed annular windows. This distinction keeps the global pin out of the local decorrelation estimate.

Lemma 3.1. *For a sufficiently separated family of annular windows at geometrically separated radii, or sufficiently separated equal-size windows on a line, local events $F_1, \dots, F_m$ satisfy*

$$\mathbb{P}\Big[\bigcap_{j=1}^m F_j\Big] \le \prod_{j=1}^m \mathbb{P}[F_j]^{1/2}\,. \tag{3.1}$$

The separation constants depend only on the fixed window shapes and d.

Proof. Identify the covariance norm of zero-mass observations with the homogeneous $H^{-d/2}$ norm. For an inner window of radius r and an outer annular window of radius R, subtract a constant in the inner variable from the logarithmic covariance kernel. After rescaling the two windows to fixed size, its derivatives of any fixed finite order are $O(r/R)$. Sobolev duality, using an order larger than $d/2$ in each variable, bounds the operator correlation between the Gaussian subspaces by Cr/R. For two equal-size windows at distance Ar, subtract the constant and both one-variable terms. The analogous bound is CA^{-2}. Smooth test functions are dense in the observation spaces, so these bounds extend to their closures.

Choose the geometric ratio, or the equal-size separation, so that the sum of the off-diagonal operator norms in each row is at most one. The joint covariance is then bounded above by twice its block diagonal. The Gaussian Hölder inequality [8, Theorem 1(i)], first for finite-dimensional projections and then by conditional expectation, proves (3.1). □

A consequence used repeatedly is the following covering estimate. At each tested scale, take centers in a fixed sufficiently fine multiple of $r\mathbb{Z}^d$. If a local event has failure probability at most p_0, then, except on an event of probability at most

$$C(1+W)^d e^{2dV} p_0^{cV}, \tag{3.2}$$

every point of B_W lies in the central tenth of a good ball of radius in $[e^{-2V}, e^{-V}]$. To see this, first take an $e^{-2V}/100$ net of B_W. Around each net point choose one nearby tested center at each of cV

separated radii. The associated windows are contained in separated annular envelopes around that net point. Apply Lemma 3.1, then sum over the net. Enlarging the central core covers the whole ball. Choosing p_0 sufficiently small makes (3.2) exponentially small in V, up to the displayed volume factor.

The physical circle averages are controlled separately. For at most Ce^{Cn} tests at radii between e^{-Cn} and e^{Cn}, with centers in $B_{e^{Cn}}$, Gaussian tails and a union bound give

$$\max |h_r(z)| \le C'n \tag{3.3}$$

except on an event of probability at most Ce^{-cn}, if C' is large enough. Consequently (1.7) gives

$$e^{-C\tau n} \le S_\eta(r,z)/S_\xi(r,z) \le e^{C\tau n}\,. \tag{3.4}$$

Neither (3.3) nor any event involving a global geodesic is inserted into the local Gaussian event.

3.2 Local witnesses and measurable geodesics

Use the fixed annulus and its inner compact subannulus

$$\mathcal{A} = \{1/2 < |z| < 7/2\}, \qquad \mathcal{K} = \{3/4 \le |z| \le 1\}\,.$$

On a fixed-probability reference event from Proposition 2.1, there is a deterministic $\delta_0 > 0$ such that, whenever $u, v \in \mathcal{K}$ and $|u - v| \le \delta_0$,

$$D(u,v;\mathcal{A}) = D(u,v), \qquad E(u,v;\mathcal{A}) = E(u,v)\,. \tag{3.5}$$

More precisely, the internal competitor is shorter than the cost of reaching $\partial\mathcal{A}$. Hence these equalities persist under arbitrary changes of the field outside $\mathcal{A}$. This is a first-exit comparison, not a claim of locality for unrestricted distances.

The ambient comparisons and the unfavorable crossing law use the radius-ten pin. Favorable trials and the site units in the field-change catalogue use their unit-radius pin. Both have centered log unit distance by (1.5); whenever a local test is rescaled, its own pin is subtracted. Its observation window includes that pin sphere. Formula (1.7), with the corresponding radius, accounts for the conversion to physical units.

All geodesics used in the counting argument are chosen by one fixed Borel rule, invariant under multiplying the metric by a constant. Such a rule can be constructed by choosing the lexicographically first exact midpoint, recursively at dyadic times. To verify measurability, on a compact proper metric ball approximate the midpoint constraints by rational points and tolerances $1/j$, then successively minimize the coordinates. The minima converge to the exact coordinate minima of the compact midpoint set. Iteration gives a Borel path at dyadic times; continuity gives the whole path. The construction is unchanged by scalar multiplication. The required metric class is Borel: symmetry and triangle inequalities can be tested on rational points; separation is tested by minima on a countable family of separated compact sets; properness by divergence of distances outside integer balls; and the length property, in the proper class, by rational approximate midpoints. Thus the selector may be extended by a fixed path off this class.

We use local favorable and unfavorable arc events. A favorable witness is an E-geodesic in $\mathcal{A}$, lying in $\mathcal{K}$, with

$$E(u,v;\mathcal{A}) \le cD(u,v;\mathcal{A})\,. \tag{3.6}$$

An unfavorable witness is a D-geodesic with the reversed strict inequality. For these crossing events, require a crossing of the two boundary spheres of $\mathcal{K}$. On the reference event, subdividing

such a crossing by a fixed number of equal own-length pieces yields a short retained witness of Euclidean diameter at most $\delta_0/2$, length in a fixed interval $[\ell_0, \ell_1]$, and endpoint separation at least $w_* > 0$. The retained piece need not cross both spheres. At least one piece retains the relevant inequality: use additivity in the geodesic metric and the triangle inequality in the other metric. It is this short piece, with its internal distances identified by (3.5), which is used in the geometric construction and endpoint net. The spatial bounds give fixed bi-Hölder moduli for its constant-speed parameterization. These local witness events are Borel. Indeed, the admissible arcs form a compact set by the two moduli; the distance, containment, and arclength conditions are closed conditions, with a countable union for strict inequalities. Projection from this compact witness space preserves Borel measurability.

3.3 Sewing comparisons along a geodesic

Here is the deterministic part of the amplification. The clipped intervals at the ends of a geodesic must be charged separately, since they need not contain an annular crossing.

Lemma 3.2 (Sewing)**.** *Suppose the central cores of good balls, with radii in $[r_-, r_+]$, cover a primary geodesic $P : [0, \ell] \to \mathbb{R}^d$. Assume their primary crossings are at least mS_D, their relevant primary and secondary connectors are at most MS_D and MS_E, and $S_E/S_D \le L$. Then the secondary endpoint distance is at most*

$$2(M/m)L\ell + CHr_+^\sigma . \tag{3.7}$$

Suppose in addition every complete primary crossing contains a block of primary length at least mS_D, with a secondary connector of cost at most b times the length of that block. If a previous global comparison is $E \le CD + e$, then a new comparison along P is

$$E(P(0), P(\ell)) \le \left(b + q_*(C - b) + \frac{e}{B_-} \right) \ell + Ce_+ + e , \tag{3.8}$$

provided $C \ge b$, where $q_* < 1$ is fixed,

$$B_- = H^{-1}(r_-/4)^\beta, \qquad e_+ = 2H(2r_+)^\sigma .$$

All paths and distances in the local hypotheses can be internal distances in their inspection annuli.

Proof. Take the preimage components of the covering balls which contain visits to their central cores. A finite minimal interval subcover has multiplicity at most two. In its order along P, consecutive balls overlap and neither contains the other, so their boundary spheres intersect. This elementary fact holds for every $d \ge 2$. Connect along the corresponding spheres and use a mandatory primary crossing to charge each complete interval. The multiplicity-two bound gives (3.7); attaching the two endpoints costs at most CHr_+^σ.

For the refined estimate, discard the two clipped end intervals. Their total primary length is at most e_+. Each remaining interval contains a useful block of primary length at least mS_D, whereas its whole length is at most MS_D. Two-color the interval family and retain one color. The retained disjoint blocks have total length at least $\theta(\ell - e_+)$, where $\theta = m/(2M)$; take $q_* = 1 - \theta$. Each block has primary length at least B_-. Use its local secondary connector and the old comparison on the complementary primary subsegments. There are at most $\ell/B_- + 1$ such complementary pieces. The old additive errors therefore total at most $e\ell/B_- + e$. Summing the primary lengths proves (3.8). □

Proposition 3.3 (Annular amplification)**.** *There are fixed constants $C_b > 1$, $p_0 > 0$, and $M, C, c > 0$ with the following property. Suppose a local event of probability at least $1-p_0$ supplies the secondary connector bound bD on every primary thin crossing, in its own metric units, where $C_b^{-1} \le b \le C_b$. Then*

$$\mathbb{P}\big[E \not\lesssim (b+\delta)D + e \text{ on } \overline{B}_R^2\big] \le C_R \delta^{-M} e^{-c\delta^M n} \tag{3.9}$$

whenever $0 < \delta, e < 1/2$ and

$$n \ge C\delta^{-M}\big(1 + \log(1+R) + \log e^{-1} + \log \delta^{-1}\big), \qquad \tau n \le c\delta\,. \tag{3.10}$$

The statement permits one of ξ, η to be zero. Without the local connector assumption, there is a rough comparison $E \le C_b D + e$ and its reverse with failure probability at most q, provided

$$n \ge C\big(1 + \log(1+R) + \log e^{-1} + \log q^{-1}\big), \qquad \tau n \le c\,. \tag{3.11}$$

Proof. For the rough comparison, confine both metrics' relevant geodesics in B_W by relative escape, taking W polynomial in R, q^{-1}. Choose a polynomial upper quantile H for the spatial constants. Use (3.2) at radii between e^{-2un} and e^{-un}, with fixed small $u > 0$, for the reference crossing and connector events. Their marginal failure can be chosen as small as needed. Equations (3.3)–(3.4) control the ratio of physical units. Lemma 3.2 then gives both rough comparisons, with endpoint errors at most $CHe^{-\sigma un}$. This proves (3.11).

For amplification, choose $N = \lceil C \log \delta^{-1} \rceil$, a large fixed L, and

$$a_j = \tfrac{1}{4} L^{-j}, \quad 1 \le j \le N, \qquad a_* = a_N \ge c\delta^{M_0}\,.$$

Use coverings with radii $[e^{-4a_j n}, e^{-a_j n}]$, in this order from smaller to larger scales. Choose confinement and spatial cutoffs $W = C_R e^{wa_* n}$ and $H = C_R e^{ha_* n}$. First choose L so that $4\beta/L < \min(\sigma/4, c_1/16)$, where $e^{-c_1 n}$ is the initial rough additive error; then choose $h > 0$ small and $w > 0$ sufficiently small relative to h, so that $Aw < ph/2$ in (2.1). The failures of confinement, spatial bounds, all coverings, and the separate pin cap sum to at most $C_R \delta^{-M} e^{-c\delta^M n}$.

On the complementary event, write $b_{\mathrm{phys}} = be^{C\tau n}$. Starting from the rough coefficient, apply (3.8). As long as the coefficient is at least b_{phys}, the coefficient and additive error satisfy

$$\begin{aligned} C_j &= b_{\mathrm{phys}} + q_*(C_{j-1} - b_{\mathrm{phys}}) + e_{j-1}/B_j, \\ e_j &= e_{j-1} + C_{j-1}E_j, \\ B_j &= H^{-1}(e^{-4a_j n}/4)^{\beta}, \qquad E_j = 2H(2e^{-a_j n})^{\sigma}. \end{aligned}$$

If the coefficient is already smaller, retain that comparison. The induction $C_j \le C_b + 1$ closes since the total additional coefficient error is small. In fact $e_j \le CHe^{-\sigma a_j n}$, and the first quotient is bounded by $CHe^{-(c_1 - 4\beta a_1)n}$, while every later quotient is at most

$$CH^2 e^{-(\sigma a_j - 4\beta a_{j+1})n}\,.$$

Their sum is at most $\delta/8$ under (3.10), after increasing M, C. The initial coefficient discrepancy is reduced by $q_*^N \le c\delta$. The pin error is $b_{\mathrm{phys}} - b \le \delta/8$, and $e_N \le e$. This proves (3.9). The proof uses the Euclidean metric without change when a parameter is zero. □

4 A routing construction in every dimension

The field modification must connect the two ends of a favorable arc to two prescribed ports, without allowing a path to bypass the intervening gates. Existence of disjoint paths alone is not enough. We need quantitative clearance, a finite deterministic catalogue, and polynomial bounds for the Cameron–Martin norms. We obtain all of these without a distinction between $d = 2$ and $d \ge 3$.

4.1 Clearance and disjoint access paths

Let Q be a fixed rational box. Its two ports are distinct fixed rational points in the relative interiors of two faces. An arc $\zeta : [0,1] \to Q$ is controlled if it stays a fixed positive distance from ∂Q and, for fixed $A < \infty$ and $0 < \alpha, \alpha' \le 1$,

$$|\zeta(s) - \zeta(t)| \le A|s-t|^{\alpha}, \qquad |s-t| \le A|\zeta(s) - \zeta(t)|^{\alpha'}. \tag{4.1}$$

Write $\Gamma = \zeta([0,1])$ and $F^{+r} = \{x : \operatorname{dist}(x,F) \le r\}$.

Lemma 4.1 (Quantitative clearance)**.** *There are $c, C > 0$, depending only on the control data and d, such that any point of Q at distance at least u from Γ can be joined to ∂Q by a polygonal path with clearance at least $cu^{1/(\alpha\alpha')}$ from Γ. Apart from a short initial attachment when the starting point is not a grid point, the path may be chosen on a rational grid of comparable mesh and with at most Cu^{-C} edges. The same statement holds for every nontrivial subarc with the inherited control data.*

Proof. The inverse parameter map on Γ is α'-Hölder. Extend it by the McShane formula for the metric $|x-y|^{\alpha'}$, and truncate the extension F to $[0,1]$. The map $P = \zeta \circ F$ is a retraction onto Γ. For $\operatorname{dist}(z,\Gamma) \le s \le 1$,

$$|z - P(z)| \le Cs^{\alpha\alpha'}. \tag{4.2}$$

Indeed, compare $F(z)$ to the parameter of a nearest arc point and use both moduli. Linear interpolation from z to $P(z)$, followed by contraction along the arc, contracts Γ^{+s} inside $\Gamma^{+Cs^{\alpha\alpha'}}$.

Take a finite union K_0 of closed grid cubes with

$$\Gamma^{+r} \subset K_0 \subset \Gamma^{+C_d r}.$$

Every bounded component of its complement is contained in $\Gamma^{+Cr^{\alpha\alpha'}}$. To prove this, take a point x in such a component. The oriented boundary of the component, viewed as a cubical $(d-1)$-cycle, has degree one about x. This follows by summing the oriented boundaries of its cubes; interior faces cancel, and a small sphere about x has degree one. The boundary is supported on K_0. If $x \notin \Gamma^{+Cr^{\alpha\alpha'}}$, the contraction above contracts that cycle in $\mathbb{R}^d \setminus \{x\}$, contradicting invariance of degree. One can first choose x off the grid and then use closedness to include the other points. This is the winding-number argument in dimension two and the same degree argument in all higher dimensions.

Choose $r = cu^{1/(\alpha\alpha')}$ so that the filled set is contained in $\Gamma^{+u/2}$. The starting point belongs to the unbounded complementary component of K_0, and therefore has a path to infinity outside K_0. Stop after reaching the boundary of Q. Clamping any earlier excursion to Q does not reduce clearance below the fixed distance of Γ from that boundary. Approximate on a grid with mesh less than $r/(100\sqrt{d})$ and erase loops. Clearance is at least $r/2$, and the number of grid vertices in Q is $O(r^{-d})$. Rationally rounding r changes only constants. The argument for a subarc is identical. □

Lemma 4.2 (Two access paths)**.** *For every sufficiently small $t > 0$, a controlled arc has two simple disjoint rational polygonal access paths from the two ports to within $C_d\nu$ of its endpoints, with either pairing allowed. Here $\nu \ge ct^C$ is a rational grid scale. The paths have at most $C\nu^{-d}$ edges, their mutual separation and the separation of nonincident edges are at least $c\nu$, and each path stays at distance at least $K\nu$ from the opposite truncated arc, for any fixed K selected before the refinement. The constants may depend on K. Outside the endpoint-tail neighborhoods, both paths have the same $K\nu$ clearance from the whole arc. Coordinates lie on a grid of mesh comparable to ν^2. A sufficiently thin tube around either path can leave Q only near its own port.*

Proof. Apply Lemma 4.1 to $\zeta([t,1])$ and to $\zeta([0,1-t])$. The opposite endpoint is at distance at least $(t/A)^{1/\alpha'}$ from each truncated arc. We obtain preliminary endpoint paths with clearance $R_0 \geq ct^C$ from the respective opposite tails.

Choose a coarse cubical mesh $q \geq ct^C$ so small that the union of all cubes meeting Γ^{+10q}, after filling every bounded face-connected complementary component, satisfies

$$\Gamma^{+10q} \subset K \subset \Gamma^{+R_0/100}. \tag{4.3}$$

This follows from the degree argument in Lemma 4.1. The unselected exterior cubes form a face-connected collection inside Q: each belongs to the unbounded exterior collection, and the fixed free collar of ∂Q joins any portions of an exterior path lying outside Q.

Refine each coarse cube into m^d cubes, with fixed m sufficiently large relative to K and d, and put $\nu = q/m$. The exterior has clearance $10q$ from the arc, and the prefixes have clearance at least $R_0/2$ from their opposite tails; both exceed $2K\nu$ with these choices. Round each preliminary endpoint path to this fine grid, and stop it at its first boundary vertex of K. Its interior lies in K, and, by (4.3) and its clearance from the opposite tail, it lies near its own endpoint tail $\zeta([0,t])$ or $\zeta([1-t,1])$. These two tails have a fixed positive separation for $t \leq 1/4$. Hence the two prefixes are disjoint.

Let G be the fine-grid graph in the union of the closed exterior coarse cubes. Deleting any one vertex leaves G connected. Within one coarse cube this follows by rerouting through a two-dimensional grid face whenever a grid path meets the deleted vertex; this uses $d \geq 2$. Neighboring coarse cubes share at least two fine-grid vertices in a common face. Their connected grids therefore remain connected to one another after the deletion. Face connectivity of the coarse exterior collection proves the assertion for G.

There are two ports and two distinct prefix exit vertices. No single vertex separates these two sets, so the set version of Menger's theorem [9, Theorem 3.3.1] gives two vertex-disjoint paths between them. Either matching is permitted. Concatenate with the prefixes and erase loops. The graph is a Cartesian grid, so disjoint nonincident edges have separation at least ν. The number of edges is bounded by the number of grid vertices, at most $C\nu^{-d}$.

Finally move every free vertex slightly toward the center of Q, by a rational homothety of displacement less than $\nu/100$, keeping the ports fixed. Separations remain at least $c\nu$. Free vertices have distance at least $c\nu$ from the boundary, and the first edge at a port points quantitatively inward. This gives the tube assertion. A rational homothety with denominator $O(\nu^{-1})$ places the new coordinates on a mesh comparable to ν^2. All clearances lose only fixed factors. Reversing ζ exchanges the endpoint labels and preserves every required metric inequality, so the free matching causes no change in later uses. □

4.2 A finite catalogue of discounted networks

We give the metric form of the geometric construction. The collector is

$$\mathcal{C} = \{3/2 \leq |z| \leq 5/2\} \Subset \mathcal{A}.$$

The source reference event fixes primary internal diameter and collar-crossing bounds A_0, a_{out}, and a primary spatial constant. Fix $0 < \varepsilon_0 \ll b_{\mathrm{trig}} \ll a_{\mathrm{out}}$. A finite embedded cubical tree in the collector is chosen so that each point of $\mathcal{C}$ is within primary distance ε_0 of a vertex. The enlarged star of each vertex, including its incident reserved edge boxes, has primary diameter at most ε_0. This is possible by (2.2), choosing the mesh after the fixed reference constant.

Each reserved edge box contains a finite set of small, separated trial annuli. At an accepted trial, there is a favorable secondary arc with source endpoints u, v and

$$L_D = D(u,v) = D(u,v;\mathrm{trial}), \qquad d_0 \le L_D \le D_0, \qquad E(u,v;\mathrm{trial}) \le (1+4v_0)L_D\,. \tag{4.4}$$

The constants d_0, D_0 and the bi-Hölder control of the arc are fixed; $0 < v_0 < 1/100$ is the accuracy. Equality of global and trial distances is secured by (3.5). In particular it holds for every admitted exterior field, not just for the field in which the witness was first found.

Proposition 4.3 (Finite forcing catalogue)**.** *Suppose $a \le \xi, \eta \in J$, and every reserved edge box has a trial satisfying (4.4). There is a deterministic finite catalogue of nonnegative smooth functions f_ℓ, supported in $\operatorname{int}\mathcal{C}$, and Borel source-validity events, with*

$$\log J_{v_0} \le C v_0^{-C}, \qquad \|f_\ell\|_{\mathcal{H}}^2 \le C a^{-2} v_0^{-C}, \qquad \|f_\ell\|_\infty \le C a^{-1}\log(v_0^{-1})\,. \tag{4.5}$$

At least one label is valid for every such source field. There is $c_{\mathrm{seed}} \ge c v_0^C$ such that the following holds for a valid label. Let a primary geodesic for the modified field $h - f_\ell$ have endpoints outside B_4. If its first and last collector visits u_0, v_0 satisfy

$$D_h(u_0, v_0; B_4) \ge b_{\mathrm{trig}}\,, \tag{4.6}$$

then it contains a block of primary modified length at least c_{seed} with a secondary modified connector of cost at most

$$(1+6v_0)e^{C\Omega} \quad \textit{times its primary modified length}, \qquad \Omega = \tau\|f_\ell\|_\infty\,. \tag{4.7}$$

Its full first-to-last collector interval stays in B_4. The statements scale in the source primary units. A source-valid modified network has primary diameter at most $3\varepsilon_0$.

Proof. Step 1: gates, paths, and labels. Apply Lemma 4.2 in each reserved box, choosing its clearance multiple larger than $100C_d$, where C_d below controls the gate radius and endpoint rounding. Choose a tail parameter $t_0 = c v_0^p$, with p larger than 1 and $1/(\alpha\sigma)$. The two endpoint tails are $[0, 2t_0]$ and $[1-2t_0, 1]$. Choose the routing mesh $\nu \ge c v_0^C$ so small that $\nu \ll v_0^{2/\sigma}$ and the clearances in Lemma 4.2 hold for these tails. Put $g = C_d\nu$, $\omega = c\nu^4$, and $\mu = c\omega^2$, with constants in that order.

Label the μ-cubes covered and hit by the full arc and by its two endpoint tails. Both conditions are required: covering provides control of the whole witness, and hitting lets a cell center be joined back to the witness. The chamber is the union of closed ω-balls about the full-arc cell centers. Each gate is the union of closed g-balls about its tail-cell centers. If T_i is the corresponding tail, then

$$T_i^{+(g-C_d\mu)} \subset G_i \subset T_i^{+(g+C_d\mu)}, \qquad \Gamma \subset \text{chamber} \subset \Gamma^{+2\omega}\,.$$

Inside a gate, join a point radially to a hit-cell center by a geometric sequence of balls, and then to the arc. Equation (2.2) bounds this part by CHg^σ. The endpoint tail adds at most $2t_0\ell_1$ in its own metric units. Primary gate diameters are at most $CH(t_0^\alpha + g)^\sigma$. Our choices make each of these errors at most $v_0 d_0/1000$, after the fixed trial-unit conversion.

Thicken the access paths by ω. Away from its own gate, each access path has polynomial clearance from the chamber; it has clearance from the opposite gate everywhere. Distinct paths and nonincident edges remain separated. Lemma 4.2 handles the portions near the box boundary. Consequently, after deleting the gates, the only incidences are those of the prescribed tree with

each reserved edge replaced by a path–chamber–path connection. The original middle segment of every reserved edge is deleted. Retaining that segment would create a bypass and would invalidate the conclusion.

All nonincident parts have separation at least $\delta_{\mathrm{sep}} \geq c v_0^{C}$. A chain of hit cells and of balls along the access paths bounds the primary diameter of the entire undiscounted network by $A_{\mathrm{net}} \leq C v_0^{-C}$. This uses spatial connectors, not the primary length of the secondary arc. There are at most $C\mu^{-d}$ relevant cells and $C\nu^{-d}$ path edges, and their rational coordinates lie on a polynomial mesh. Thus the number of labelled possibilities satisfies the first bound in (4.5). Restrict the deterministic list to labels having the stated separation and incidence properties. Every source witness produces one of these labels. Compactness of the controlled witness space and the covered-and-hit conditions make source validity Borel.

Step 2: the smooth discount. Choose a polynomially small $\theta \in (0,1)$ so that

$$\theta A_{\mathrm{net}} \ll \varepsilon_0, \qquad \theta A_{\mathrm{net}} \ll H^{-1}\delta_{\mathrm{sep}}^{\beta}. \tag{4.8}$$

Let U be the closed network just constructed, including its gates, and choose a still smaller polynomial collar width $s > 0$. If φ is a fixed smooth mollifier supported in the unit ball, set

$$f = \xi^{-1} \log(\theta^{-1}) \big(\mathbf{1}_{U^{+s/4}} * \varphi_{s/4} \big). \tag{4.9}$$

It equals $\xi^{-1} \log \theta^{-1}$ on U, is nonnegative, and is supported in $V = U^{+s}$. Its primary Weyl weight is θ on U and lies in $[\theta, 1]$ everywhere. The ratio of secondary and primary weights lies in $[e^{-\Omega}, e^{\Omega}]$. Make s small enough that all full-density collar attachments have cost at most $v_0 \theta d_0/1000$ and that V retains the separation and incidence properties.

The Cameron–Martin norm is a constant multiple of the homogeneous $H^{d/2}$ norm. For the fixed bounded support, bound it by an inhomogeneous Sobolev norm of integer order at least $d/2$. Derivatives of the mollified indicator are bounded by powers of s^{-1}. Since every geometric scale is polynomial in v_0, this proves the last two bounds in (4.5). Under translation and dilation the homogeneous $H^{d/2}$ norm is unchanged. The support avoids both the unit and radius-ten pin spheres.

Step 3: forcing a chamber crossing. A network competitor has primary modified cost at most $3\varepsilon_0$. The unchanged outer collars therefore keep the first-to-last collector interval of the modified geodesic inside B_4, and indeed in the fixed inspection annulus. An excursion outside V whose endpoints are separated by $\delta_{\mathrm{sep}}/2$ has primary cost at least $H^{-1}(\delta_{\mathrm{sep}}/2)^{\beta}$. The network shortcut is cheaper by (4.8); such an excursion cannot occur.

Suppose no chamber is visited through both of its enlarged opposite gates. Delete, in the finite incidence tree, a gate not visited by the path in each chamber. Every resulting component either lies in one enlarged vertex star or is contained in a single isolated chamber. The separation just proved prevents a path from moving between different such components without visiting a deleted gate. The modified distance between the two collector endpoints is at most $3\varepsilon_0$, by joining each endpoint to the network and then using its discount. If the geodesic never visits V, that entire segment has its source length, which already contradicts (4.6). Otherwise all its visits to V lie in one enlarged star. Before the first and after the last such visit, the field is unchanged. Join these two visits through the corresponding source star; the s-collar costs are absorbed by its diameter bound. The resulting source path has cost less than $7\varepsilon_0$ and stays in B_4. This again contradicts (4.6). Hence both gates of some chamber are visited.

Take the subpath between those visits. Since the primary weight is everywhere at least θ, its length is at least $\theta(L_D - v_0 d_0/100)$, and hence at least $\theta d_0/2$. A secondary connector follows

the favorable arc, with the gate and collar attachments already estimated. The favorable arc and gate portions receive the discount; the outer collar errors were bounded at full density. The connector costs at most $\theta e^{C\Omega}((1+4v_0)L_D + v_0 d_0/100)$ plus the full-density collar error, while the primary block costs at least $\theta(L_D - v_0 d_0/100)$. With the preceding slack these bounds give (4.7). Set $c_{\mathrm{seed}} = \theta d_0/2$. This proves the proposition. □

5 A quantitative two-parameter comparison

The geometric construction produces a cheap connector when a geodesic visits a prescribed site. We next show that a geodesic with a persistently bad comparison must visit too many such sites. The count is made for finitely many deterministic shifts; no random Cameron–Martin shift is used.

5.1 Favorable arcs have fixed positive probability

Lemma 5.1. *For each sufficiently small $v > 0$, if $\tau \le cv^C$, a fixed reference trial contains a favorable secondary arc satisfying (4.4), after a fixed subdivision and unit conversion, with probability at least a fixed $p_* > 0$. The constants p_*, d_0, D_0 and the arc moduli do not depend on v.*

Proof. The centered variable

$$Y = \log E(0,e_1) - \log D(0,e_1)$$

has a uniformly bounded second moment. Cantelli's inequality therefore gives

$$\mathbb{P}[E(0,e_1) \le (1+v)D(0,e_1)] \ge cv^2\,. \tag{5.1}$$

Choose a fixed local reference event with failure much smaller than p_0. If the favorable-arc event at coefficient $1+2v$ had probability less than p_*, the complement on this reference event would give reverse connectors, along every secondary thin crossing, of primary cost less than $(1+2v)^{-1}$ times secondary length. Proposition 3.3 would then give

$$D \le \big((1+2v)^{-1} + cv\big)E + cv^{A+1}$$

at the unit pair, outside probability $cv^2/4$, for $\tau \le cv^C$. Choose A so large that the inverse point-distance moments give $\mathbb{P}[D(0,e_1) < v^A] \le cv^2/4$. Intersecting with (5.1) contradicts $D \le (1-c'v)D$. The constants c in the error terms are chosen small before the depth. This proves a fixed positive lower probability. The fixed subdivision in Section 3.2 gives the controlled short arc and its fixed length and separation bounds. □

Choose finitely many sufficiently separated small trials in each reserved tree-edge box. Lemma 3.1 applied to their failures makes the probability that a box has no favorable trial as small as desired. Their number and scale are fixed, since p_* is fixed. Separately bound their finitely many pin differences by a fixed Gaussian quantile. For $\tau \le cv^C$, the conversion to site units changes $1+2v$ to at most $1+4v$. Intersecting with the primary reference event yields source-valid forcing sites with arbitrarily high fixed probability. All tests remain local and constant-invariant.

5.2 A finite change-of-measure inequality

Lemma 5.2. *Let $f_1, \dots, f_N$ be deterministic Cameron–Martin shifts with norm at most M. Suppose Borel events A_i, B satisfy*

$$\sum_{i=1}^{N} \mathbf{1}_{A_i}(h) \ge L\mathbf{1}_B(h), \qquad \sum_{i=1}^{N} \mathbf{1}_{A_i}(h+f_i) \le U\,.$$

Then

$$\mathbb{P}[B] \le \frac{U}{L} \min\{N, \exp(M\sqrt{2\log N})\}\,. \tag{5.2}$$

Proof. Cameron–Martin change of measure gives

$$L\mathbb{P}[B] \le U\,\mathbb{E} \max_i \exp\{-W(f_i) - \|f_i\|_{\mathcal{H}}^2/2\}\,.$$

The expectation is at most N. For $p > 1$, its L^p bound is $N^{1/p}\exp((p-1)M^2/2)$. Optimize over p, or use $p \downarrow 1$ if the optimum is less than one, to obtain the displayed bound. This proof needs no independence among the shifts. □

5.3 Improving a comparison coefficient

The next lemma records the only costly part of the comparison. Its parameter dependence is exponential in a polynomial, rather than a tower of exponentials.

Lemma 5.3 (Local improvement)**.** *Fix the guard ball B_{R_s} sufficiently large. There are fixed $q_0, p_0 > 0$ and a polynomial P with positive coefficients such that the following holds. Let $0 < a \le \xi, \eta \in J$, $0 < v < 1/100$, and $1 + 20v \le c \le C_b$. There are $g, \varepsilon_*, t_* > 0$, which may be chosen independently of c in this range, satisfying*

$$e^{-P(a^{-1},v^{-1})} \le g \le v, \quad \log \varepsilon_*^{-1} \le P(a^{-1}, v^{-1}), \quad t_* \ge e^{-P(a^{-1},v^{-1})}, \tag{5.3}$$

such that, if $\tau \le t_*$ and

$$\mathbb{P}[E \le cD + \varepsilon_* \text{ on } \overline{B}_{R_s}^2] \ge 1 - q_0, \tag{5.4}$$

then the local unfavorable primary-arc event at coefficient $c - g$ has probability at most $p_0/2$.

Proof. Step 1: source references and deterministic site catalogues. Choose common source references so that D has a fixed spatial upper bound on the guard ball, a fixed internal competitor of cost U_0 between points of B_2, and an unchanged barrier of cost greater than $2U_0$ outside B_4. These events have arbitrarily high fixed probability. All field modifications below are nonpositive and supported strictly inside B_4, so the barrier confines every relevant modified geodesic. The ambient field has the radius-ten pin. In the same-radius site catalogue below, however, every site unit is $S_D(r, z) = r^{\phi(\xi)}e^{\xi h_r(z)}$, using the unit-radius pin at that site. This distinction is needed for invariance of the units under the other site shifts.

Fix first $K_0 > 2\beta + 3$ and a net exponent $D_{\mathrm{net}} > 2d(K_0 + 3)/\sigma + 3$. Choose $\mu > D_{\mathrm{net}} + 2$. Only then choose the fixed number of favorable trials and the fixed local reference quantiles to give the covering failure Δ^μ below. These choices alter numerical constants but not K_0 or D_{net}. Let $\Delta > 0$ be small. Use radii $r_j = \Delta e^{-\lambda j} \in [\Delta^2, \Delta]$ and centers in $c_g r_j \mathbb{Z}^d$, with fixed small c_g and large λ. From (3.2) and the forcing-site construction, except on an event of probability at most Δ^μ, the central cores of valid sites cover the source region. The fixed single-site failure probability is chosen after μ, as just specified. Separately impose the pin cap $|h_{r_j}(z)| \le C \log \Delta^{-1}$. The physical unit errors are $O(\tau \log \Delta^{-1})$.

Fix endpoints $x, y \in B_2$ with $|x - y| \ge w_*$, and follow the selected source primary geodesic through a fixed radial annulus around x. Visits at radial levels separated by 10Δ yield at least $c\Delta^{-1}$ sites with disjoint full inspection balls. Spatial upper and lower bounds and the fixed local reference crossings give, with the preselected K_0,

$$2\Delta^{K_0} \le S_D(r, z) \le \Delta^{-K_0}\,. \tag{5.5}$$

Color the center lattice by residues modulo a fixed integer so that inspection balls of one color and radius are disjoint. Pigeonhole over radius, color, and dyadic unit bins. For small Δ there is a catalogue containing at least

$$N_{\mathrm{site}} = \lceil \Delta^{-1/2} \rceil$$

valid visited sites of one radius and color, with primary units in one deterministic bin $[Q, 2Q)$. The number of radius–color–bin catalogues is $O((\log \Delta^{-1})^2)$, and each has at most $C\Delta^{-K_{\mathrm{cat}}}$ candidate centers. Include all these centers in the deterministic catalogue; validity and the actual unit bin are source-event conditions, not a random choice of shifts.

At a valid visited site, the source crossing costs a fixed multiple of its unit, while a network replacement costs at most $3\varepsilon_0$ times that unit. Thus replacing any k chosen source crossings saves at least $a_0 kQ$, for fixed $a_0 > 0$. Here one charges a chosen inward crossing from radius $5r/2$ to $3r/2$ immediately before a core visit, not the whole first-to-last source visit interval. These chosen crossings lie in the disjoint inspection balls, so their parameter intervals are disjoint.

Step 2: many admissible shifts at a bad source. Let $\kappa > 0$ be a small fixed constant and $m_0 = \lceil \kappa k \rceil$. Index shifts by labelled subsets Z of the chosen deterministic catalogue, with $m_0 \le |Z| \le k$ and at most one label at each site. Set $f_Z = \sum_{z \in Z} f_z$. For fixed k,

$$N \le (CJ_v \Delta^{-K_{\mathrm{cat}}})^k, \qquad \|f_Z\|_{\mathcal{H}} \le kM_0, \qquad M_0^2 \le Ca^{-2}v^{-C}\,. \tag{5.6}$$

The supports avoid the global pin sphere and every site pin sphere in this catalogue. Hence all primary units, and the ratio of primary and secondary units, are unchanged by these shifts.

The bad source event B includes the common references, the supply of visited sites, the comparison $E \le cD + \eta_{\mathrm{loc}}$, and

$$E(x,y) \ge (c - g_\Delta) D(x,y), \qquad D(x,y) \le U_0\,. \tag{5.7}$$

Let A_Z additionally require that all labels in Z are source-valid, have units in $[Q, 2Q)$, and the selected primary geodesic in $h - f_Z$ satisfies the source trigger at every site of Z.

At each valid visited site choose its least valid catalogue label. Starting with any k such sites, apply their selected discounts. If the selected modified geodesic fails a trigger, restore that site's field and use the short source connector supplied by the failure of (4.6). This costs at most $2b_{\mathrm{trig}}Q$. Recompute the selected geodesic and continue, removing the least-index failing site. The procedure terminates. It loses at most $2b_{\mathrm{trig}}kQ$ of the original saving $a_0 kQ$. Restoring each remaining site's whole collector interval costs at most $2A_0 Q$. Restore in chronological order from the first visit to the last visit of that site; overlapping intervals only reduce the number of replacements. Since a fully restored path cannot beat the source distance, at least κk sites remain, if b_{trig} and then κ were chosen small enough. The terminal subset satisfies A_Z.

A terminal subset of size at least m_0 is contained in at most $N_{\mathrm{site}}^{k-m_0}$ of the initial subsets. Since $\binom{N_{\mathrm{site}}}{k} \ge (N_{\mathrm{site}}/k)^k$, we obtain

$$\sum_Z \mathbf{1}_{A_Z}(h) \ge N_{\mathrm{site}}^{m_0} k^{-k} \mathbf{1}_B(h)\,. \tag{5.8}$$

Step 3: few shifts at a common output. Fix an output field h. Every active event $A_Z(h + f_Z)$ uses the same selected output primary geodesic. Site units coincide across all active events, since none of the shifts changes any site pin. At each active physical site, Proposition 4.3 supplies a block of output primary length at least $c_{\mathrm{seed}}Q$ and a secondary shortcut with coefficient $(1 + 6v)e^{C\Omega}$, where $\Omega = \tau \max_Z \|f_Z\|_\infty$. Full collector intervals stay in their disjoint inspection balls, so blocks at distinct sites have disjoint parameter intervals.

Fix one active source $H_0 = h + f_{Z_0}$ and discard the at most k sites in Z_0. The other shortcuts are unchanged at this source. If m remain, apply the source comparison only at H_0, on the complementary output geodesic pieces. Restore the supports in Z_0 at total primary cost at most $2A_0kQ$. If $L_i \geq c_{\mathrm{seed}}Q$ are the retained block lengths, the resulting competitor gives

$$E_{H_0}(x,y) \leq cD_h(x,y) - \big(c - (1+6v)e^{C\Omega}\big)\sum_{i=1}^{m} L_i + 2cA_0kQ + (m+2k+1)\eta_{\mathrm{loc}}\,. \tag{5.9}$$

No comparison is asserted at any other hybrid field. The near-bad condition at H_0, and monotonicity under its positive restoration, give the lower bound $(c - g_\Delta)D_h(x,y)$.

Choose, with sufficiently small fixed constants,

$$g_\Delta = c_1 v c_{\mathrm{seed}}\Delta^{K_0}/U_0, \qquad \eta_{\mathrm{loc}} \leq c_2 v c_{\mathrm{seed}}\Delta^{K_0}, \tag{5.10}$$

$$\Omega \leq c_3 \min\{v, a_0\Delta^{K_0}/U_0, vc_{\mathrm{seed}}\Delta^{K_0}/(C_bU_0)\}. \tag{5.11}$$

Then (5.9) implies $m \leq K_1(v)k$, with $K_1(v) \leq Cv^{-C}$. Thus there are at most $CK_1(v)k$ active physical sites. The number of labelled subsets is at most

$$\sum_{Z} \mathbf{1}_{A_Z}(h + f_Z) \leq (CK_1(v)J_v)^k\,. \tag{5.12}$$

Step 4: the probability bound and the local event. Apply Lemma 5.2 using (5.6), (5.8), and (5.12). With $\ell = \log\Delta^{-1}$, it gives

$$\log\mathbb{P}[B] \leq -\tfrac{1}{2}\kappa k\ell + k\log k + k\log(CK_1J_v) + kM_0\sqrt{2k(K_{\mathrm{cat}}\ell + \log(CJ_v))}\,. \tag{5.13}$$

First choose k large enough for any desired fixed power of Δ. Then choose ℓ larger than a polynomial in a^{-1}, v^{-1}. The negative linear term dominates the square-root term, giving $\mathbb{P}[B] \leq \Delta^{2\mu+2}$. Sum over the logarithmically many site catalogues.

Finally round the endpoints of a local unfavorable arc to a spatial net. Their separation is at least w_*, and (3.5) identifies internal and global distances. The primary spatial bound and the source comparison control the rounding in both metrics, so a mesh of size a fixed power of g_Δ preserves (5.7) with a smaller fixed multiple of g_Δ. Its cardinality is at most $\Delta^{-D_{\mathrm{net}}}$, with the exponent chosen in Step 1. Indeed a mesh proportional to $g_\Delta^{1/\sigma}$ suffices; the extra powers of v^{-1} and the fixed reference constants are absorbed by making $\log\Delta^{-1}$ larger than a polynomial in v^{-1}. Thus the order of choices is K_0, D_{net}, then μ and the reference data, then k, and finally Δ. In particular there is no circular choice between a net exponent and the single-site failure probability. The failures of the common source references, the site supply, and (5.4) are charged once, outside this endpoint union bound. Their probabilities are chosen small fixed fractions of p_0. A local unfavorable event of probability greater than $p_0/2$ would therefore force a bad source event of probability bounded below by a fixed positive constant, contradicting (5.13).

Take $\Delta = e^{-P_0(a^{-1},v^{-1})}$, with P_0 large enough for all preceding requirements. Set g to the smaller fixed multiple of g_Δ used in rounding, choose $\varepsilon_* \leq \eta_{\mathrm{loc}}$, and choose t_* to satisfy the favorable-arc, physical-unit, and (5.11) conditions. Equations (4.5) and (5.10) show that these choices satisfy (5.3), after enlarging one polynomial P. This proves the lemma. □

5.4 The increment estimate

Proposition 5.4. *Fix J and R. There are $B, M, C, c > 0$, a fixed guard radius R_s, and fixed $c_*, p_s > 0$ such that, with*

$$X_h^\xi = \log(1 + \rho_h^\xi),$$

we have

$$\mathbb{P}[\|X_h^\xi - X_h^\eta\|_R > \delta] \le C\delta^{-M} \exp(-c\delta^M n) \tag{5.14}$$

whenever $0 < \delta < 1/2$,

$$n \ge C\delta^{-M} \log(C/\delta), \qquad |\xi - \eta| n \le c\delta, \tag{5.15}$$

and either of the following conditions holds:

(i) *$a \le \xi, \eta \in J$ for some $a > 0$, and*

$$|\xi - \eta| \le \exp(-Ca^{-B}\delta^{-B}); \tag{5.16}$$

(ii) *the external seed*

$$\mathbb{P}[\|\rho_h^\xi - \rho_h^\eta\|_{R_s} > c_*\delta] \le p_s \tag{5.17}$$

holds. In this alternative, one of the parameters may be zero.

All constants are chosen on J before introducing a.

Proof. We first obtain (5.17) at any prescribed accuracy $u > 0$ from the stronger spacing $\tau \le \exp(-Ca^{-B}u^{-B})$. Start with the rough comparison in Proposition 3.3, on a sufficiently large fixed guard ball and with failure q_0. Apply Lemma 5.3, then Proposition 3.3 with coefficient error $g/2$ and additive error ε_*. This returns a global comparison with coefficient reduced by $g/2$ and the same failure bound q_0. Repeat until the coefficient is at most $1 + 20v$.

This is an iteration of deterministic implications between probability bounds, not an intersection of successful random events. Thus the failure probability is not multiplied by the number of iterations. All depths and the number of iterations are bounded by $\exp(P(a^{-1}, v^{-1}))$. The same single parameter-spacing requirement makes every depth compatible with $\tau n \le c\delta$. Repeat with the metrics interchanged. A fixed upper quantile for both diameters on the guard ball, and v a sufficiently small fixed multiple of u, give

$$\mathbb{P}[\|\rho_h^\xi - \rho_h^\eta\|_{R_s} > u] \le p_s \quad \text{if} \quad \tau \le \exp(-Ca^{-B}u^{-B})\,. \tag{5.18}$$

All finite polynomial powers have been absorbed by increasing B, C. The values g, ε_*, t_* used in this iteration are uniform in the current coefficient c; this is why the additive tolerance can be held fixed throughout.

It remains to amplify an external seed; this proves both alternatives. Intersect the seed with a fixed local reference event for which (3.5) holds. A primary thin crossing has length in a fixed interval $[m, L]$. Divide it into a fixed number N of equal primary-length pieces so that each has Euclidean diameter at most $\delta_0/2$. At their endpoints, the seed and (3.5) imply a secondary internal distance at most the primary piece length plus $c_*\delta$. Summing gives a secondary connector of cost at most $(1 + Nc_*\delta/m)$ times the primary crossing length. Choose c_*, p_s sufficiently small relative to the local reference constants and p_0. Proposition 3.3, in both orientations, gives outside the probability in (5.14)

$$E \le (1 + \delta/2)D + \delta/2, \qquad D \le (1 + \delta/2)E + \delta/2\,.$$

These inequalities imply $|\log(1 + E) - \log(1 + D)| \le \delta$. The conditions reduce to (5.15), after increasing M, C. Using (5.18) with $u = c_*\delta$ proves the generic alternative. The seeded argument never divides by a parameter, so it includes zero. □

6 Construction of the common continuous family

We use Proposition 5.4 first on compact positive parameter intervals and then near zero. The logarithmic profiles avoid any preliminary upper bound on a whole parameter family. They are only interpolated as functions; we do not assert that an interpolant is a metric.

6.1 Positive parameter intervals

Fix $I = [a, b] \Subset (0, \xi_c)$ and $R < \infty$. Choose $D > \max(B, 1)$, a sufficiently large constant A, and

$$k_j = \lceil A 2^{Dj} \rceil, \qquad w_j = (b-a)2^{-k_j}, \qquad \delta_j = 2^{-j-6}. \tag{6.1}$$

Let G_j be the relative dyadic grid of level k_j in I. Compare every point of G_{j+1} with both bracketing points of G_j, using depth $n_j = \lfloor c_1 \delta_j / w_j \rfloor$ with c_1 sufficiently small. A large enough A makes the generic seed and depth conditions valid for all j. The total failure probability at level j is at most

$$C 2^{k_{j+1}} \delta_j^{-M} \exp\left(-c \frac{\delta_j^{M+1}}{w_j} \right). \tag{6.2}$$

This is summable. Indeed, the magnitude of the negative exponent is a fixed multiple of $2^{k_j - (M+1)j}$, whereas the logarithm of the other factors is $O(k_{j+1} + j)$.

Let X_j be the piecewise linear interpolant of the Banach-valued profiles X_h^ξ, $\xi \in G_j$. On a successful level, comparison with both brackets gives $\sup_{\xi \in I} \|X_{j+1}^\xi - X_j^\xi\|_R \le 2\delta_j$. Borel–Cantelli gives an almost surely uniformly Cauchy sequence. Its limit X_I is continuous in ξ with values in $C(\overline{B}_R^2)$. Set $\rho_I = e^{X_I} - 1$. At every point in the nested union of grids it agrees with the original metric. The dense grid and joint continuity then imply the triangle inequality, symmetry, and vanishing diagonal at every parameter. Thus ρ_I is initially a continuous family of finite pseudometrics.

For each deterministic $\xi \in I$, it is a version of the prescribed metric. To see this, fix $\delta > 0$ and take $n = \lfloor c\delta / |\xi - \eta| \rfloor$ in Proposition 5.4. As $\eta \to \xi$, its conditions eventually hold and its failure bound tends to zero. This proves stochastic continuity of the original logarithmic profiles. Grid values approaching ξ converge almost surely to X_I^ξ and in probability to the prescribed X_h^ξ, so the two agree almost surely. Exponentiation gives the same assertion for ρ.

Carry this out for countably many compact intervals with rational endpoints and all integer R. On overlaps the families agree on a countable dense set of deterministic parameters, hence everywhere by continuity. All grids are deterministic and countable. Their Banach-valued interpolation limits, the overlap tests on countable dense sets, and the exceptional event are therefore measurable functions of h; evaluation of the continuous limit also gives joint measurability in the parameter. Give the family the Euclidean value on the exceptional set. This constructs a field-measurable continuous pseudometric family for $0 < \xi < \xi_c$.

6.2 Continuity at zero

We now work with the positive-parameter version just constructed. At each deterministic parameter it has the canonical law. The endpoint argument uses only Lemma 2.2, not a rate in that lemma.

Fix the guard radius in Proposition 5.4. By convergence in probability, there is a deterministic function $\omega(a) \downarrow 0$ such that, for all sufficiently small a,

$$\sup_{0 \le \xi \le 2a} \mathbb{P}[\|\rho_h^\xi - \rho_h^0\|_{R_s} > c_0 \omega(a)] \le p_s/4. \tag{6.3}$$

For example, choose decreasing thresholds 2^{-k}, and then decreasing deterministic values of a for which the defining limit gives this estimate. This is a supremum of individual probabilities, not a probability of a random supremum. Choose c_0 small relative to c_*. Put

$$\kappa = \frac{1}{2(M+1)}, \qquad \varepsilon_a = \max\{\omega(a), a^\kappa\}\,. \tag{6.4}$$

By the triangle inequality through ρ^0, every deterministic pair in $[0, 2a]$ has the external seed at tolerance $\varepsilon_a/8$.

Apply the seeded increment estimate to $(\xi, 0)$ with $n = \lfloor c\varepsilon_a/\xi \rfloor$. Its depth condition holds because

$$\frac{\varepsilon_a^{M+1}}{\xi} \geq ca^{-1/2} \gg \log \varepsilon_a^{-1}\,.$$

Consequently,

$$\sup_{0<\xi\leq 2a} \mathbb{P}[\|X_h^\xi - X_h^0\|_R > \varepsilon_a/8] \leq Ca^{-\kappa M} e^{-ca^{-1/2}}\,. \tag{6.5}$$

The polynomial floor in (6.4) is what makes this estimate useful without a rate for ω.

For dyadic a, choose

$$D > \max\{B(1+\kappa), 1\} + 2, \qquad L > D + \kappa M + 3\,. \tag{6.6}$$

The fine grids in $[a, 2a]$ have relative levels and tolerances

$$k_j(a) = \lceil Aa^{-D} 2^{Dj} \rceil, \qquad \delta_j(a) = \varepsilon_a 2^{-j-6}, \qquad w_j(a) = a2^{-k_j(a)}\,. \tag{6.7}$$

Take $n_j = \lfloor c\delta_j/w_j \rfloor$. The generic seed condition follows from

$$a^{-B}\delta_j^{-B} \leq Ca^{-B(1+\kappa)} 2^{Bj} \leq c\, k_j(a)$$

for sufficiently large A and small a. The depth condition also holds. The sum of the failures of all successive two-bracket comparisons is bounded by

$$\sum_{j\geq 0} C2^{k_{j+1}(a)} \delta_j(a)^{-M} \exp\left(-c\frac{\delta_j(a)^{M+1}}{w_j(a)}\right) \leq C\exp(-c2^{ca^{-D}})\,. \tag{6.8}$$

For the last inequality, the negative exponent is bounded below by $ca^{-1/2}2^{k_j(a)-(M+1)j}$, and dominates $Ck_{j+1}(a) + Cj + C\log a^{-1}$ uniformly in j.

It remains to reach the first fine grid without paying its cardinality using a marginal estimate. Take a relative dyadic base grid with spacing comparable to $c\varepsilon_a a^L$, contained in the first fine grid. It has at most $Ca^{1-L-\kappa}$ nodes. The union bound using (6.5) costs at most

$$Ca^{1/2-L} e^{-ca^{-1/2}}\,. \tag{6.9}$$

Now compare each node of the first fine grid with one of its base-grid brackets. Use the external seed, tolerance $\varepsilon_a/8$, and depth $n = \lceil a^{-L} \rceil$. The spacing ensures the drift condition. The union bound is at most

$$C\varepsilon_a^{-M} \exp(CAa^{-D} - c\varepsilon_a^M a^{-L}) \leq C\exp(-ca^{-(L-\kappa M)})\,. \tag{6.10}$$

This is the only intermediate comparison. It is valid because $L - \kappa M > D$.

All three failure bounds are summable over dyadic a. Off these failures, the first fine grid is within $\varepsilon_a/4$ of X^0, and the successive fine comparisons add at most $2\sum_j \delta_j(a)$. The union of the grids is dense in the slab. For a point on a finer grid, repeatedly choose its bracketing ancestor in the preceding grid; summing the corresponding δ_j bounds its deviation from the first grid. Positive-parameter continuity therefore gives

$$\sup_{\xi\in[a,2a]} \|X_h^\xi - X_h^0\|_R \le C\varepsilon_a \tag{6.11}$$

for all sufficiently small dyadic a, almost surely. Since $\varepsilon_a \to 0$, exponentiating proves continuity at zero. Intersect over integer R. We have obtained a field-measurable continuous family of finite pseudometrics on $[0, \xi_c)$.

7 Simultaneous metric properties and normalization

The remaining argument is independent of the quantitative comparison. It excludes exceptional degenerate parameters and transfers the fixed-parameter properness and geodesicity to the entire continuous family.

7.1 No random degenerate parameter

Choose a countable additive subgroup $\mathcal{H}_0$ dense in the Cameron–Martin space, consisting of bounded smooth pinned functions. Such a group is obtained from rational linear combinations of a dense collection of smooth Cameron–Martin representatives, subtracting their sphere averages. Choose a countable dense set $\mathcal{Q}$ of positive parameters containing the grid parameters used above. At these parameters the modification agrees with the canonical metric.

By Weyl scaling and Cameron–Martin absolute continuity, on a common full-probability event, for $f \in \mathcal{H}_0$ and $\xi \in \mathcal{Q}$,

$$e^{-\xi\|f\|_\infty}\rho_h^\xi \le \rho_{h+f}^\xi \le e^{\xi\|f\|_\infty}\rho_h^\xi\,. \tag{7.1}$$

Include all $\mathcal{H}_0$-translates of the event on which the continuous family exists. Continuity extends (7.1) to every parameter. The resulting event can be chosen exactly $\mathcal{H}_0$-invariant by intersecting all its countably many translates.

Fix a compact parameter interval I and two disjoint compact balls A, B with rational data and positive separation. The function

$$F_h(\xi) = \min_{x\in A,\, y\in B} \rho_h^\xi(x,y)$$

is continuous. Its zero set $Z(h) = \{\xi \in I : F_h(\xi) = 0\}$ is a measurable random compact set. Measurability follows by expressing its hits on open intervals through minima on a countable compact exhaustion. On the invariant full-probability event, (7.1) gives $Z(h+f) = Z(h)$ for all $f \in \mathcal{H}_0$. Set $Z = \varnothing$ outside this event, so this invariance holds everywhere.

If Y is the indicator of a basis hitting event of Z, Cameron–Martin change of measure gives

$$\mathbb{E}\left[Y \exp\{W(f) - \|f\|_{\mathcal{H}}^2/2\}\right] = \mathbb{E}Y, \qquad f \in \mathcal{H}_0\,. \tag{7.2}$$

Continuity in L^2 extends the identity to every $f \in \mathcal{H}$. Gaussian exponentials span a dense subspace of Gaussian L^2, so Y is almost surely constant. There are only countably many basis hitting events, and they determine a closed subset of I. Thus $Z(h)$ is almost surely equal to one deterministic compact set S.

If $S \neq \varnothing$, choose a deterministic $\xi_* \in S$. At this parameter, the family agrees almost surely with a canonical metric, or with the Euclidean metric if $\xi_* = 0$. Its minimum on $A \times B$ is strictly positive, contradicting $F_h(\xi_*) = 0$. Hence $S = \varnothing$. Intersect over a countable exhaustion of parameter intervals and all separated rational compact-ball pairs. Every two distinct spatial points belong to such a pair. Therefore every member of the continuous family is a genuine metric, on one event.

7.2 Common barriers at infinity

Use the radius-ten pin and the measure-preserving dilation

$$(Th)(z) = h(2z) - h_{20}(0)\,.$$

It is invertible on pinned fields. On a common event containing all integer iterates of T, exact scaling at the dense canonical parameters extends, by continuity, to

$$\rho_h^{\xi}(2^j x, 2^j y) = 2^{j\phi(\xi)} e^{\xi h_{10\,2^j}(0)} \rho_{T^j h}^{\xi}(x, y) \tag{7.3}$$

for every ξ and integer j.

For a compact $I \Subset [0, \xi_c)$, joint continuity and nondegeneracy imply

$$W_I(h) := \min_{\xi \in I} \rho_h^{\xi}(\partial B_1, \partial B_2) > 0\,.$$

Poincaré recurrence applied to the sets $\{W_I \geq 1/k\}$ shows that for almost every h, some $k < \infty$ has $W_I(T^j h) \geq 1/k$ at arbitrarily large j. The spherical averages satisfy $\operatorname{Var} h_{10\,2^j}(0) = O(j+1)$. Gaussian tails and Borel–Cantelli give $h_{10\,2^j}(0)/j \to 0$. We do not use Brownian increments of spherical averages, which would be a dimension-specific assertion. Equation (7.3), with $\min_I \phi > 0$, therefore gives

$$\min_{\xi \in I} \rho_h^{\xi}(\partial B_{2^j}, \partial B_{2^{j+1}}) \longrightarrow \infty \tag{7.4}$$

along a sequence $j \to \infty$.

For the dense canonical parameters, every path from B_{2^j} to $B_{2^{j+1}}^c$ crosses the two spheres. Hence its endpoint distance is at least their metric separation. This inequality passes to every parameter by continuity, including continuity of the infimum over the two compact spheres. The barriers thus confine bounded metric balls uniformly over I in Euclidean bounded sets.

Metric continuity implies that Euclidean convergence implies metric convergence. Conversely, a metric-convergent sequence is bounded in metric distance and therefore Euclidean bounded by the barriers. Every Euclidean subsequential limit has zero distance from the metric limit, so nondegeneracy makes it the same point. The two topologies coincide. A closed bounded metric ball is Euclidean closed and bounded, hence compact in both topologies. All metrics are proper, simultaneously.

7.3 Geodesics and the median convention

Fix ξ, endpoints x, y, and canonical parameters $q_n \in \mathcal{Q}$ tending to ξ. Their distances between x, y are bounded. Parameterize canonical geodesics at constant relative arclength on $[0, 1]$. The common barriers confine their images in a fixed Euclidean compact set K. For every $\varepsilon > 0$,

$$\min\{\rho_h^{q}(u, v) : q \in I,\ u, v \in K,\ |u - v| \geq \varepsilon\} > 0$$

by compactness and nondegeneracy. The geodesics are therefore equicontinuous in Euclidean distance. Arzelà–Ascoli gives a uniformly convergent subsequence. Joint continuity passes the constant-speed distance identity to its limit, which is a ρ_h^ξ-geodesic. This is a deterministic argument on the selected event, so it applies to all parameters and all endpoints without a further uncountable intersection.

It remains to return to the unit-sphere pin and the point-median convention. Equation (1.5) multiplies the family by a positive parameter-continuous factor and preserves all conclusions. For this pin, put

$$\beta(\xi) = \operatorname{med} \rho_h^\xi(0, e_1), \qquad \beta(0) = 1\,. \tag{7.5}$$

We claim that β is positive and continuous. At any positive parameter the law of the unit distance has full support in $(0, \infty)$. To see this, choose a bounded smooth Cameron–Martin function f equal to one on a tube about $[0, e_1]$, and adjust its sphere average to zero with a bump near a disjoint part of the pin sphere. Weyl scaling makes $t \mapsto \log \rho_{h+tf}^\xi(0, e_1)$ continuous. A competitor in the tube sends it to $-\infty$ as $t \to -\infty$; a mandatory initial crossing in a ball on which $f = 1$ sends it to $+\infty$ as $t \to +\infty$. Each open distance interval is therefore hit at some rational t. Cameron–Martin equivalence implies that the original unit-distance law gives positive mass to every such interval.

Consequently the median is unique and has strict distribution-function inequalities on either side. Convergence in probability of the continuous metric family at a deterministic parameter implies convergence of these medians. At zero the unit distance converges in probability to the constant one, which also forces its medians to converge to one. This proves the claim; an atomlessness argument is unnecessary.

Define, for the unit-sphere-pinned field,

$$\widetilde{D}_h^{\gamma(\xi)} = \beta(\xi)^{-1} \rho_h^\xi\,. \tag{7.6}$$

At every deterministic positive parameter, (1.4) and the original normalization give $\beta(\xi) = e^{-m(\xi)}$. Thus (7.6) is a modification of the originally prescribed median-one metric, not a new deterministic normalization. The positive continuous factor β^{-1} preserves continuity, properness, geodesicity, and the topology. Finally compose with the homeomorphism $\gamma \mapsto \xi$. This proves Theorem 1.1.

A Sources and scope of the unification

This proof combines the planar manuscript [12] and the higher-dimensional manuscript [11], rather than assuming either continuity theorem. The common one-metric estimate is imported at the exact strength stated in Proposition 2.1. The probabilistic comparison is reconstructed in Sections 3 and 5. The routing argument in Section 4 replaces both dimension-specific access-path constructions. All changes from planar grids to cubical grids incur only fixed dimension-dependent powers in the catalogue size and Sobolev norm.

The new proof of Lemma 2.2 uses the geodesic occupation-energy idea appearing in the supplied first-order manuscript [13, Section 3.2], but it uses only the already uniform one-metric estimates and Gaussian Poincaré. Neither its first-order conclusion nor its entropy bootstrap is imported. The three-grid construction at zero uses only convergence in probability. The common-event proof includes nondegeneracy, properness, geodesicity, the pin change, and the exact median normalization. No $O(\gamma \log(1/\gamma))$ rate, continuity at $\gamma = \sqrt{2d}$, uniformity in the dimension, or Lean verification of this manuscript is asserted.

For reproducibility, the source revisions read for this unification were the 46-page planar PDF and the 34-page higher-dimensional PDF supplied with the conversation. The latter consists of a

25-page main text and its uniform-estimate appendix. In particular, its Appendix A is part of the source, not an additional continuity input.

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[10] Supplied project manuscript. *Higher-dimensional exponential metrics: A proposed axiomatic existence and uniqueness proof.* File `higher-dimensional-lqg-metric-revised.pdf`, in the supplied October 17–19, 2026 archive. Theorems 1.2 and 1.4, Corollaries 10.2 and 10.4, and Lemma 10.3. This strong characterization is the standing higher-dimensional input retained from [11]; it is not re-proved here.

[11] Supplied project manuscript. *Almost sure continuity of the higher-dimensional LQG metric.* File `higher_dimensional_lqg_continuity.pdf`, 34-page revision supplied for comparison. Lemma 2.1 and Appendix A are the one-metric inputs; Sections 3–5 are the comparison and common-family source arguments.

[12] Supplied project manuscript. *Almost sure parameter continuity of the planar LQG metric.* File `lqg_continuity_d2_complete_proof.pdf`, 46-page revision supplied on October 25, 2026. Section 2, particularly Proposition 2.8, supplies the uniform planar metric estimates. Sections 3–5 are the geometric, comparison, and common-family source arguments.

[13] Supplied project manuscript. *First-order fluctuations of small-parameter exponential metrics.* File `unified-first-order-exponential-metrics.pdf`, in the supplied October 17–19, 2026 archive. Only the occupation-energy idea in Section 3.2 is used; its first-order theorem is not an input.